

How We Created the Future Projections

Now that we have a model that has learned how climate and terrain relate to the pika's presence today, we want to use it to look ahead: **given future climate change, how would each site's suitability for the pika change by around 2050?** To answer this, we feed the model the climate expected in the future in place of today's. Here is how we produced the future projections for each of our features:

Climate Features Projection

- **bio1** — mean annual temperature
- **bio4** — temperature seasonality
- **bio5** — max temperature of the warmest month
- **bio6** — min temperature of the coldest month
- **bio12** — annual precipitation
- **bio15** — precipitation seasonality

For these features we used [WorldClim CMIP6](#) projections, the standard source for downscaled future climate. Specifically, we chose:

- **Mild Scenario (SSP2-4.5):** it imagines a future where the world takes some action on emissions, so they rise a bit longer and then stabilise around the middle of the century, leading to moderate warming.
- **Extreme Scenario (SSP5-8.5):** it imagines a future with little to no action, where fossil-fuel use keeps growing throughout the century, pushing warming to the highest levels among the standard scenarios.

Deriving Temperature-Based Features

- max_temperature
- winter_max_temperature
- summer_min_temperature
- winter_min_temperature
- months_above_25

The model includes also several temperature-based features.

Future projections for these variables are available from [TerraClimate](#), but they define future climate differently. TerraClimate uses fixed global warming levels of +2°C and +4°C above pre-industrial conditions, while our WorldClim projections refer to specific emissions scenarios at around 2050: SSP2-4.5 and SSP5-8.5. Since a +2°C world does

not exactly correspond to SSP2-4.5 in 2050, combining the two sources would mix incompatible future conditions. To ensure consistency, we therefore derived these features from the WorldClim projections.

For each site, we first measured the projected change in temperature between the present and future climate. We then applied this change to the corresponding temperature-based feature, using summer warming for summer variables and winter warming for winter variables.

This adjustment was calculated separately for each site.

Estimating the Snow Features

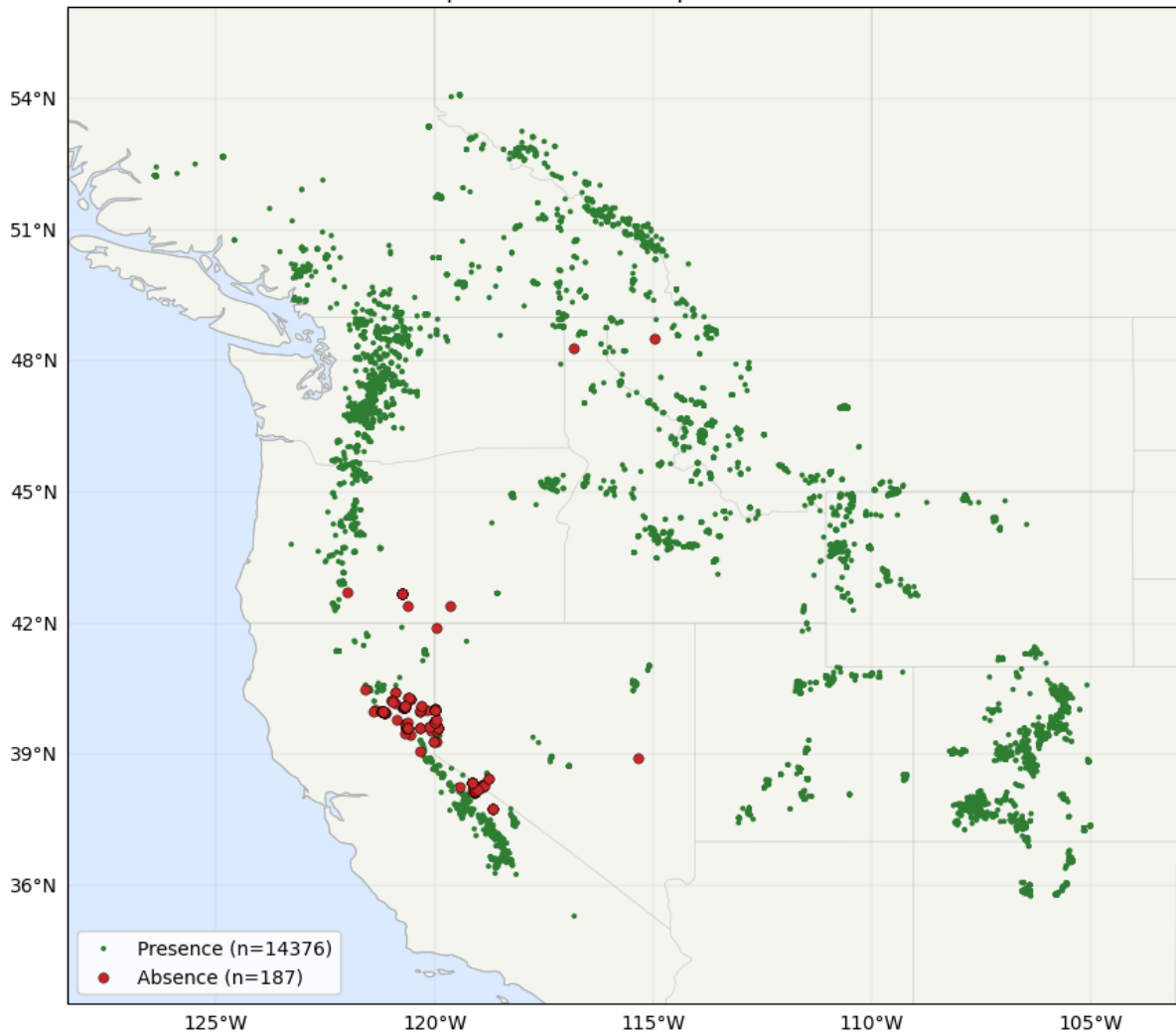
- months_with_snow
- estim_snow
- Max_snow_YearBefore
- AnnualMean_swe
- AnnualMean_YearBefore_swe
- months_with_snow_YearBefore
- springMean_YearBefore_swe

Snow-related features were estimated using projected winter warming. Because warmer winters lead to less snow, we scaled down each site's present-day snow values according to its projected winter warming : a site whose winters warm more loses proportionally more snow.

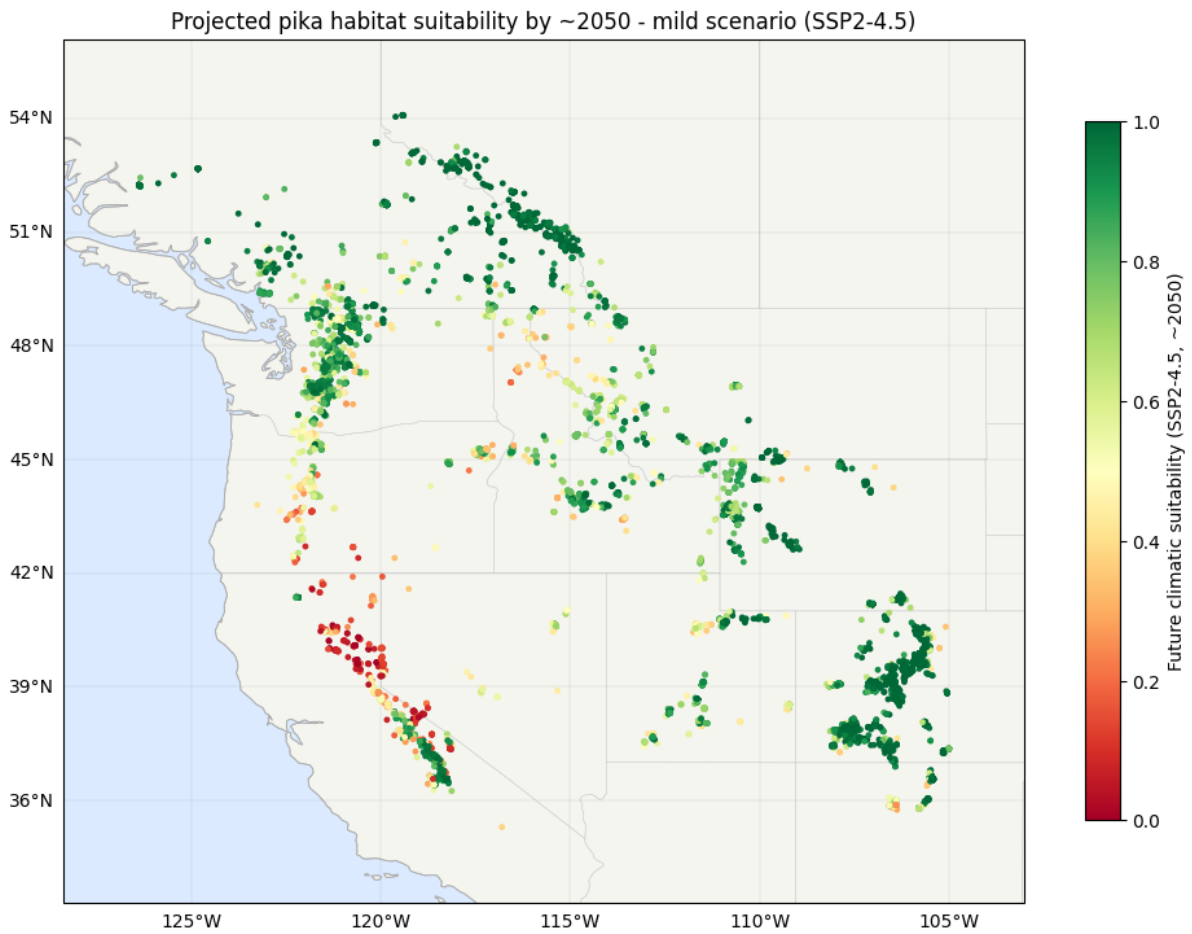
Suitability Score

Our suitability score ranges from 1 (highly suitable) down to 0 (not suitable at all).

Observed pika occurrences — presence vs absence



Mild Scenario



Mean suitability today: 0.934

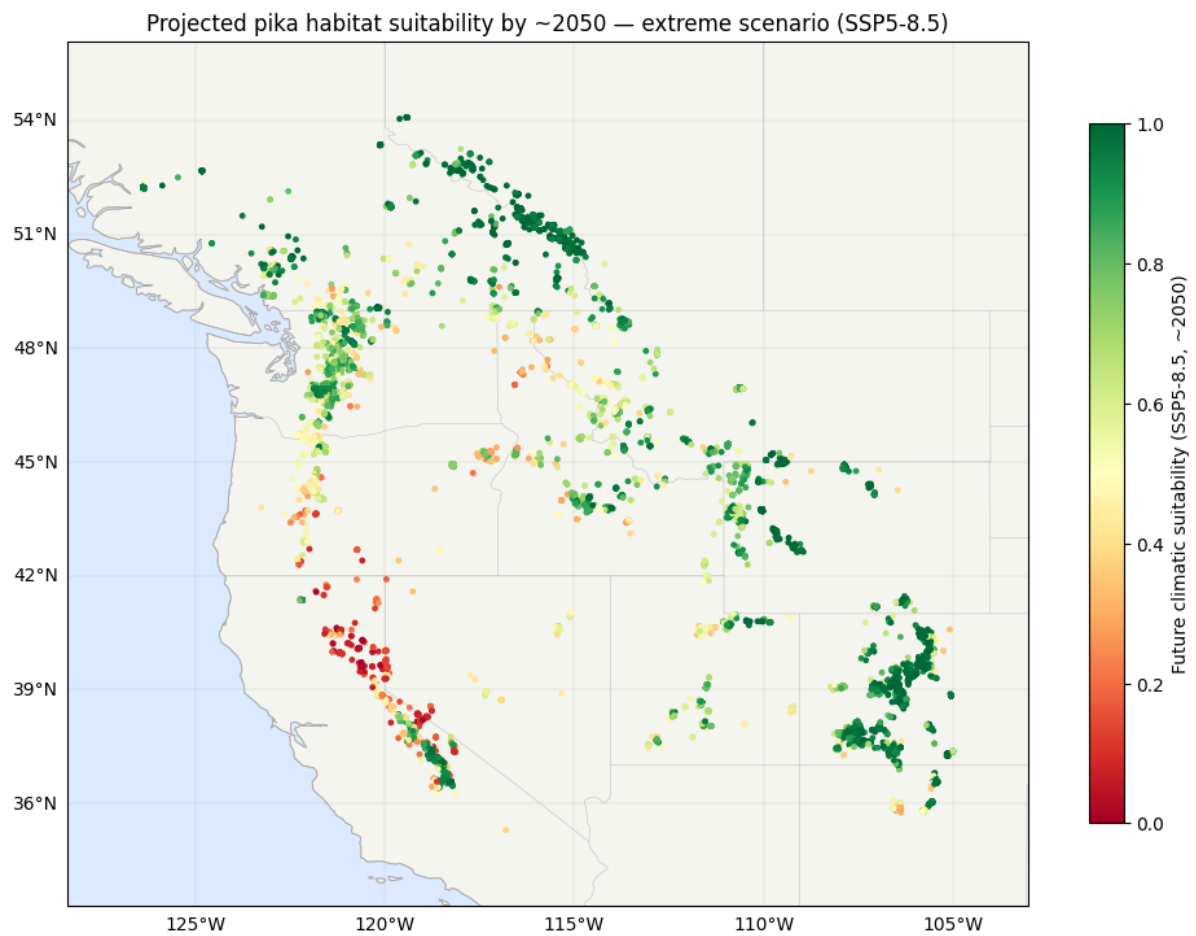
Mean suitability mild: 0.826

Mean change in suitability: -0.108

Sites declining: 12670 / 14580

Sharp decline (less than 0.20 suitability): 3203

Extreme Scenario



Mean suitability today: 0.934

Mean suitability extreme: 0.785

Mean change in suitability: -0.148

Sites declining: 13673 / 14580

Sharp decline (less than 0.20 suitability): 4729